

REVIEW ARTICLE

A Narrative Review on Surgical Site Infections and Preventive Measures

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Received: 25 July, 2025

Acceptance: 09 August, 2025

Published: 15 September, 2025

ABSTRACT

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INTRODUCTION

Surgical site infection (SSI) is defined as an infection occurring within 30 days after an operation is performed at incision or adjacent site. SSI are one of the leading cause of morbidity and mortality around the world. It is one of the common types of hospital-acquired infections in developing countries (5.6% of surgical procedures) in relation to developed countries and 3rd most common reported nosocomial infection accounting for 10-40% of all nosocomial infections.^[1,2,5] Globally, SSI rates from 2.5% in Europe to 41.9% in Africa to 0.04%-22% in India.^[3,4,6,7] A Centres for Disease Control and Prevention (CDC) survey from 2015 and reported that

approximately 110,800 patients suffered from SSIs but from 2020, there was 5% reduction in SSIs after improving surgical techniques, sterilisation of instruments, operation theatre practices.^[8,9]

Classification of SSI

SSIs are classified by using a set of standard clinical diagnostic criteria and the most commonly used one are CDC America diagnostic criteria in Table 1.^[10] According to them, superficial incisional involving skin only, deep incisional involving fascial and muscle layers or organ / space incisional involving any part of the anatomy other than incisional which was opened or manipulated during an operation.

Sl. No.	Type of SSI	Diagnostic criteria Patient has at least one of the following:
I.	Superficial incisional SSIs	1. Purulent drainage from superficial incision. 2. Organism(s) identified from an aseptically-obtained specimen from the superficial incision or subcutaneous tissue by a culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment 3. A superficial incision that is deliberately opened or re-accessed by a surgeon, physician or physician designee and culture or non-culture based testing of the superficial incision or subcutaneous tissue is not performed AND 4. Patient has at least one of the following signs or symptoms: localised pain or tenderness; localised swelling; erythema; or heat 5. Diagnosis of a superficial incisional SSI by a physician or physician designee

II.	Deep Incisional SSIs	6. purulent drainage from the deep incision that is deliberately opened(excludes any known multi-part/multi-phase procedures that occur over more than one operative episode during same admission that is documented in the medical record by a surgeon prior to first phase of the procedure, re-accessed, or aspirated by a surgeon, physician or physician designee or spontaneously dehisces AND 7. organism(s) identified from the deep soft tissues of the incision by a culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (for example, not Active Surveillance Culture/Testing [ASC/AST]) or culture or nonculture based microbiologic testing method is not performed. A culture or non-culture based test from the deep soft tissues of the incision that has a negative finding does not meet this criterion. AND 8. patient has at least one of the following signs or symptoms: fever (>38°C); localized pain or tenderness; abscess or other evidence of infection involving the deep incision detected on gross anatomical exam, histopathologic exam, or imaging test
III.	Organ/space Incisional SSIs	9. purulent drainage from a drain placed into the organ/space (for example, closed suction drainage system, open drain, T-tube drain, CT guided drainage) 10. organism(s) identified from fluid or tissue in the organ/space by a culture or non-culture based microbiologic testing method which is performed for purposes of clinical diagnosis or treatment (for example, not Active Surveillance Culture/Testing [ASC/AST]) 11. an abscess or other evidence of infection involving the organ/space detected on: gross anatomical exam or histopathologic exam or imaging test evidence definitive or equivocal for infection

Table 1. Classification of SSIs

*“Incision” refer to the incision made for primary surgical procedure and laparoscopic trocar site and or is used to place a drain at the end of procedure and term “stab wound” refers to an incision made at another site, generally to accommodate drain.

*Term physician is used in the NHSN SSI criteria may be interpreted to mean a surgeon, infections disease physician, emergency physician, other physician on the case or physician designee (nurse practitioner or physician’s assistant).

The Superficial incisional SSIs are further divided into 2 specific types:

- A. Superficial Incisional Primary (SIP)- a superficial incisional SSI that is identified in the primary incision in a patient that has had an operation with one or more incisions (for example, C-section incision or chest incision for CAGB)
- B. Superficial Incisional Secondary (SIS)- a superficial incisional SSI that is identified in the secondary incision in a patient that has had an operation with more than one incision (for example, donor site incision for CAGB)

The following do not quantify as criteria for meeting the NHSN definition of superficial incisional SSI:

1. Diagnosis/ treatment of cellulitis does not meet superficial incisional SSI criteria

2. A stitch abscess alone (minimal inflammation and discharge confined to the points of suture penetration).
3. A localized stab wound or pin site infection; depending on the depth, these infections might be considered either a skin or soft tissue.

The Deep incisional SSIs are also further divided into 2 specific types:

- A. Deep Incisional Primary (DIP) – it is identified in a primary incision in a patient that has had an operation with one or more incisions (for example, C-section incision or chest incision for CBGB)
- B. Deep Incisional Secondary (DIS) – it is identified in the secondary incision in a patient that has had an operation with more than one incision (for example, donor site incision for CBGB)

Etiology

The contaminating microorganism may be endogenous or exogenous. In majority of cases, it is caused by an endogenous infection and occurs when the incision becomes contaminated with microorganisms derived from the patient’s skin (2%) or an opened internal organ. So, optimal preoperative skin preparation is essential.^[2] The patient’s skin is the direct source of contamination in only 2% of cases, leaving 98% of cases related to airborne particles.^[3] Exogenous infection occurs when external

microorganisms contaminate the operative site during the procedure.^[11] Exogenous microorganisms are vectored by airborne particles, the staff (hand, other areas of the skin, and mucous membrane) or, more rarely, inanimate objects (instruments, material, furnishing or irrigation solutions).^[2] Sources includes surgical instrument, the theatre environment and the air. External micro-organisms can also contaminate the wound at the time of an incident, or gain access to the wound following surgery before the wound was healed.^[10]

Surgical-site contamination by airborne particles is ascribable in 30% of cases to direct settling of the particles on the wound and in 70% of cases to settling on the instruments and surgeon's hand followed by transfer to the wound.^[12] Particle size influences the tendency to settle on surfaces. Particles smaller than 5 µ remain suspended in the air, those larger than 100 µ settle rapidly, and those of intermediate size (5–100 µ) may settle on potentially contaminated surfaces then migrate to another sites. Particles may carry variable bacterial loads, depending on their source.^[13]

The most common isolated organism are staphylococcus aureus (28.2%), pseudomonas aeruginosa (25.2%), Enterococcus spp. And E. coli (7.8%), coagulate-negative staphylococci (staphylococcus epidermidis) (7.1%) and enterococcus faecalis (5.6%).^[12]

Risk Factors

There are several risk factors that can increase the likelihood of developing an SSI. These includes patient's related and surgical risk factors.

The patient's risk factors for occurrence of SSI are advance age, gender (in male 6.4% and in female 4.1%) smoking, alcoholism, comorbidities like high ASA score (3 and 4), diabetes even after controlling hyperglycaemia, obesity or BMI >25, malnutrition, immunosuppression, low serum albumin concentration, haemoglobin level <12g/dl. Other physiological states that increase SSI risk include preoperative infections, tobacco use, history of prior skin and soft tissue infections, malignancy, history of radiation therapy and chemotherapy, trauma, shock, blood transfusion, hypoxia and hemodynamic instability.^[6,7-9,14,15]

External risk factors include preoperative hospital stay (>8 days), emergency surgeries, contaminated and dirty wound, prolonged duration of operation (operative time more than 3 hours) surgeon's skill (open surgical approach), inadequacies in either the surgical scrub or the antiseptic preparation of the skin, adequacy and timing of antimicrobial prophylaxis, insertion of foreign material or implants, inadequate sterilisation of surgical instruments inadequate handling of instruments, intraoperative blood loss(>600ml), High intraoperative temperature, the presence of perioperative infection, number of people(>10) in the operative theatre, and Use of drains.^[6,7,14] For these reasons, the World Society of Emergency Surgery (WSES) developed a position paper for the prevention of SSI in the operative room (OR).^[14]

According to the presence and degree of contamination, wounds can be classified as (Table 2): clean wounds, clean-contaminated wounds, contaminated wounds, dirty or infected wounds.^[16]

Sl. No	Type of Wounds	Definition
1.	Clean wounds	An uninfected operative wound in which no inflammation is encountered and the respiratory, alimentary, genital, or uninfected urinary is not entered.
2.	Clean-contaminated wounds	An operative wound in which the respiratory, alimentary, genital or urinary tracts are entered under controlled conditions and without unusual contamination.
3.	Contaminated wounds	Open, fresh, accidental wounds. In addition, operations with major breaks in sterile technique or gross spillage from the gastrointestinal tract, and incisions in which acute, non-purulent inflammation is encountered including necrotic tissue without evidence of purulent drainage
4.	Dirty or infected	Old traumatic wounds with retained devitalized tissue and those that involve existing clinical infection or perforated viscera. The organisms causing postoperative infection were present in the operative field before the operation.

Table 2. Classification of Wounds

Complications of SSI

There are local and systemic complications of SSI (Table 3). Spontaneous wound breakdown/dehiscence is the most commonest local complication while bacteraemia with possibility of spread and sepsis are the most important systemic complications. The complications are entirely preventable if there is a way to early detect the SSI and initiate treatment as early as possible.^[17] However, a recent study indicated that about 20–60% of surgical patients develop those

preventable complications because the majority of cases occur after patients are discharged from the hospital. If not treated urgently, SSI complications often require secondary surgery and are related to a 3.2% of fatality rate.^[17,18]

According to the International Surgical Wound Complications Advisory Panel (ISWCAP) 2020 report, even if SSIs are expected to occur within 30 days after surgery, the specific time when the majority of them would develop is key to early detect

SSI and thus, to prevent and early intervene against their pressing and fatal complications. Knowing the specific time is also very important to reduce the likelihood of wound complications, such as

bacteraemia, progressing to a more complex situation.^[18]

S. No.	Complication	
1.	Local	Delayed wound healing, superimposed infections, abscess formation, & osteomyelitis
2.	Systemic	Bacteraemia, distant spread & sepsis, organ failure

Table 3: Complications of SSIs

Prevention of SSI

Multiple interventions have been proposed and employed over the past decades in an attempt to prevent SSI. These include skin cleansing protocols, hair removal, the maintenance of intraoperative normothermia, the preoperative antimicrobial prophylaxis administration, the use of plastic adhesive skin barriers, the high flow oxygen supplementation, the wound protection, the sterility of instruments, the bowel preparation, the length of the incision, and the delayed primary incision closure.^[14] The WHO's surgical safety checklist is a 19-item list to improve adherence with best practice and decrease surgical site infection incidence and to promote more consistent implementation of best practices.^[9]

1. OPERATION THEATRE QUALITY

The operation theatre is a complex structure and require Civil, mechanical, electrical, electronic and biomedical integration to match the standard of operation theatre quality and its regular maintenance.

a) Structure system

The surface of operating rooms, windows, ceiling panel, rear panels of built-in-fitting, etc., should be smooth, easy to clean, compatible with the use of chemical and physical cleaning agents, free from dust accumulation and must be waterproof, fireproof and resistant to knocks.^[19] Floors surfaces should be resistant to dampness, easy to clean, stain-resistant, pleasant for extended standing, and appropriate for wheeled traffic. The colour scheme in the operation rooms should be in accordance that little objects may be quickly located if dropped.^[20]

Special rooms should be provided for performing operations and processing tools and other objects. Functional intervention must be implemented in order to ensure that "dirty" and "clean" pathways be kept separate. The size of storage areas for dirty material, clean material, supplies, instruments, equipment and drugs must be determined in accordance with the type and volume of activity of the operating unit. In addition this, the movement and activity of the people should be controlled and patient flow is also structured and logical from entrance to exit because these can affect the number of microorganisms and therefore the risk of infection.^[21]

b) Ventilation and air delivery system

HVAC (heating, ventilation, air conditioning) systems plays a fundamental role and intended to bring forth health, comfort, and safety of occupants by maintaining thermal and air quality conditions.^[22]

Unidirectional-flow systems ("laminar airflow or LAF") enable SSIs to be contained are ventilation (dilution), air distribution, room pressurization (infiltration barrier) and filtration (contaminant removal). Low-velocity unidirectional flow tends to minimize the spread of airborne contaminants and direct them towards the exhaust outlets.^[23] Laminar airflow through HEPA filters, displays 99.97% efficiency in removing airborne particles of 0.3 μm and above, can be supplied to the operating area by ceiling-mounted (vertical flow) or wall-mounted (horizontal flow) units. Most operating theatres have conventional ventilation and laminar air-flow systems with HEPA filters are generally used for orthopaedic and other implant surgery.^[23-25]

The operating theatres should be kept at a higher pressure than in corridors and adjacent areas. This positive pressure prevents the flow of air from less sterile areas into more sterile areas.^[23]

With regard to ventilation, various international scientific organisations recommend a minimum of 15-20 air exchangers per hours and a minimum of 4 exchanges of outdoor air per hour in operating theatre.^[26,27]

c) Water system

Biofilms of water distribution systems and points of use have long been recognized as a rich environment for growth of *Legionella*, mycobacteria, *Pseudomonas*, and other waterborne organisms. Eliminating biofilms and their pathogenic residents is a major challenge, as organisms dwelling in these environments may be especially impervious to disinfectants.^[28]

There are currently no specified limitations for the bacteriological purity of the water used for surgical hand scrubs. Therefore, after using the medicated soap for surgical hand antisepsis, the hands must be rinsed with clean water. Infections clearly linked to contaminated hands of surgeons after surgical hand scrub with contaminated water have not yet been documented.^[21]

2. OPERATION THEATRE ENVIRONMENT

The operation theatre environment must be safe and healthful via controlling the pollutant sources and microenvironmental variations. Meticulous designing, maintenance, periodic inspection, further education through continuing teaching and training for paramedical healthcare workers can accomplish this.^[29]

It is recommended that all patients, staff and visitor's hand and body contact surfaces must be cleaned between each patient. It is good practices points that Cleaning and disinfecting clinical care equipment, including anaesthesia machines, after each patient, and before the next patient arrives in the operating room. The clean beds with fresh, clean linen to be brought into operating theatre complex directly from clinical areas. There is no need to place patients with suspected or confirmed contact-transmissible multi-drug-resistance bacterial infection/colonization at the end of an operating list as long as operating room is cleaned and disinfected to standard between patients, and theatre ventilation is running without interruption.^[30]

a) STAFF ATTIRE

It is recommended that do not allow scrubbed staff to wear jewellery below the elbow. When jewellery cannot be removed, the area around and underneath any item of jewellery must be carefully cleaned as much as possible during the scrubbing process. Also do not allow scrubbed and un-scrubbed staff to wear artificial or polished nails in OR. It is also good practice points to ensure that all staff working in the operating room wear a head covering and a face mask in accordance with local policies and change or cover operating theatre attire e.g. with a single-use disposable gown and change footwear if leaving the operating theatre complex with intention of returning.^[30]

b) Patient and Visitor Attire

The patients specific theatre wear must be appropriate for the procedure, clinical setting, and that provides easy access to the operative site and areas for placing devices, such as intravenous cannulas.^[31]

In current practice, the accompanying parents or carers would only be permitted to enter the anaesthetic room, not the operating room itself, and they are only allowed to do that for the shortest time possible. Thus, there is no need for them to wear scrubs or any PPE. For birthing partners of women who are undergoing caesarean procedures, or anyone else who enters the operating room itself, they may still pose very little hazard as they are most likely going to be a safe distance from the operating field.^[30]

c) Foot traffic in the operating room

7% of door openings were related to unexpected events pertaining to the surgical procedure; 26% to a need for material, indicating suboptimal preoperative

planning and preparation; and 27% to the entry of people who had no role in the surgical procedure. There is strong agreement that the number of people in the OR should be kept at 5 or 6 at the most to ensure that the airborne bacterial count does not exceed 10 CFU/m³.^[32]

d) Electronic gadgets in OR

The introduction of computers, tablets, radios and, above all, cell phones into the OR is a potential source of contamination. Among these devices, 44% to 98% carry resistant microorganisms (Gram-negative rods and *S. aureus*). Therefore, the introduction of portable electronic devices into the OR is best avoided, and when their use is mandatory, they should be thoroughly cleansed using an alcohol-related solution before they are taken into the OR.^[33]

3. PREPARATION BEFORE SURGERY

a) Temperature

Normothermia to keep core body temperatures from dropping during surgery is maintained by combinations of forced warm air, skin warming, and warmed intravenous fluids.^[9]

When temperature is below 36.0°C start active warming in the ward or in the emergency department and if temperature is equal or above 36.0°C, start active warming at least 30 min prior induction of anaesthesia. On transfer to the theatre suite the active warming should be continued or re-started as soon as possible and the patient should be encouraged to walk to the theatre where appropriate.^[34]

Patient and their families and carers should be informed that staying warm before surgery will lower the risk of postoperative complications and hospital environment may be colder than their own home so they should bring additional clothing such as a dressing gown, warm cloths to keep comfortably warm and if they feel cold at any time during hospital stay they should inform to nursing staff.^[35]

b) Antibiotic prophylaxis

The administration of antibiotic prophylaxis is recommended in all surgical site infection prevention guidelines, despite the absence of RCTs. In the absence of trial data, guideline consensus is that antibiotics should be given intravenously within 60 minutes of the incision to maximize tissue concentration of the antibiotic. However, give prophylaxis earlier for operation in which a tourniquet is used. Additional recommendations include dosing antibiotics according to the timing and pharmacokinetics for e.g. the serum half-life and necessary infusion of antibiotics, and patient's weight to ensure that adequate tissue concentrations are achieved and administering subsequent doses of antibiotics for lengthy procedures if excessive bleeding occurs. Also antibiotic treatment is given in addition to prophylaxis to patients having surgery on a dirty or infected wound.^[9,31] Although the optimal

duration of prophylactic antibiotics is not known, prolonged antimicrobial prophylaxis is increasingly associated with patient harm, such as acute kidney injury.^[36]

c) Preparation of surgical site

Preparation of surgical site in the OR makes a crucial contribution to SSI prevention. It should include at least showering with an antiseptic detergent solution on the day before and morning of operation, without hair removal, provided this measure does not interfere with surgical requirements. Use of topical chemical depilatory agents: hair removal is not among the preventive measures for SSI and should not be performed routinely; in addition, shaving with a mechanical razor should be banned.^[37] If hair removal is necessary, it should be removed in the preoperative holding area and not in the operating room.^[9]

The first choice unless contraindicated or the surgical site is next to a mucous membrane is alcohol-based solution of chlorhexidine and its alternative is aqueous solution of chlorhexidine. When chlorhexidine is contraindicated the alcohol-based solution of povidine-iodine and if both alcohol based solution and chlorhexidine are unsuitable than choice of antiseptic skin preparation is aqueous solution of povidine-iodine.

Multiple guidelines recommend that surgical site antisepsis should be performed with a product that contains alcohol and another antiseptic agent (eg, chlorhexidinegluconate or povidone iodine).^[9] Topical alcohol is highly bactericidal but does not have persistent activity when used as monotherapy for skin antisepsis. Products that combine alcohol and antiseptic agents are available in the US. Chlorhexidinegluconate plus alcohol appears to be superior to povidone iodine plus alcohol for the prevention of surgical site infections.^[38]

d) Nasal decontamination

It is recommended for nasal decolonize surgical patients with intranasal antistaphylococcal agent (mupirocin ointment or povidine iodine) and antistaphylococcal skin antiseptic agent (chlorhexidinegluconate solution or wipes) for 5 days for high-risk procedure in cardiothoracic surgeries or orthopaedic surgeries.^[39] A prospective cohort study that include 8 surgical categories (abdominal, orthopaedic, urological, neurological, cardiovascular, thoracic and plastic surgery and solid organ transport found that decolonization strategies did not reduce MRSA surgical site infections.^[40] As a result, decolonization is typically focused on orthopedic, cardiothoracic or high-risk procedures such as spine and brain surgeries. However, the precise timing, agent, and frequency of application are unclear because trials addressing this issue have used different strategies. The decolonization strategy should be completed as close to the surgical procedure as possible.^[9,39]

4. PREVENTION DURING SURGERY

a) Temperature

The patient's temperature should be monitored and recorded before induction of anaesthesia and then every 30 min until end of surgery. Induction of anaesthesia should not begin unless the patient's temperature is 36.0°C or above. The ambient temperature of the theatre suite should be at least 21°C while patient is exposed inside the theatre. A direct estimate of core temperature of the patient should be measured. These sites are pulmonary artery catheter, distal oesophagus, urinary bladder, zero heat-flux (deep forehead), sublingual, axilla, rectum. TEMPERATURE SETTING on forced-air warming devices should be set at maximum and then adjusted to maintain a patient temperature of at least 36.5°C.^[9,34,41]

Before infusion of intravenous fluids and blood products, they should be warmed to 37°C using a fluid warming device and all irrigation fluids used intraoperatively should be warmed in a thermostatically controlled cabinet to a temperature of at least 36.5°C.^[9,34]

However, the use of forced air warming mattresses has been criticised as disrupting the laminar airflow and increased airborne bacterial counts.^[42]

b) Hand decontamination

The operating team should wash their hands prior to the first operation on the list using an aqueous antiseptic surgical solution, with a single-use brush or pick for the nails, and ensure that hands and nails are visibly clean. Before subsequent operations, hands should be washed using either an alcoholic hand rub or an antiseptic surgical solution.^[31]

c) Containers and instruments must remain covered

It is recommended that for all surgical/operative procedures, lay up the instruments and prosthetic materials as close as possible to when they are needed. The duration of instrument tray opening correlates directly with the contamination rate. It is 4% after 30 min, 15% after 1 hour and 30% after 4 hours.^[43,44]

d) Drapes

Impermeable, disposable, non-woven drapes deserve preference.

Do not use non-iodophor-impregnated incise drapes routinely for surgery as they may increase the risk of surgical site infection. The operating team should wear sterile gowns in the operating theatre during the operation.^[31]

e) Gloves

A recent study found that perforated gloves doubled the risk of SSI. Double gloves reduce the risk of perforation during the procedure, but perforation still occurs 4% of the time after the procedure. However,

several reported outbreaks have been traced to contaminated hand by the surgical team, despite wearing sterile gloves.^[45]

Consider wearing 2 pairs of sterile gloves when there is a high risk of glove perforation and the consequences of contamination may be serious.^[31] The gloves should be changed before performing the incision, as 12% of gloves are contaminated after draping and 24% once patient installation is completed.^[43]

f) Diathermy

If diathermy is to be carried out than use evaporation to dry antiseptic skin preparation and avoid pooling of alcohol based preparations.^[31]

g) Optimal oxygen

Maintain optimal oxygen during surgery. During major surgery and in recovery period, the haemoglobin saturation of more than 95% should be maintained.

h) Insulin use

Routinely insulin should not be given to patients who do not have diabetes to optimise blood glucose postoperatively as a measure of reducing the risk of SSI.

i) Surgical incision dressing

Cover surgical incisions with an appropriate interactive dressing at the end of operation.^[31] The use of monofilaments has not been proven beneficial.^[33] Absence of drainage is increasingly recommended, as well as early drain removal after 24 hours.^[37] Intracavity lavage and wound irrigation should not be done to reduce the risk of SSI

j) Duration of surgery

Longer the surgical or operative time has been shown to increase the SSI risk and it occurs when operative time is above the 75th percentile.^[42,43] A longer operative time may reflect intra-operative complications or greater procedural complexity should be taken out.^[42]

k) Order of patients in the operating room:

Contaminated procedures are the best performed at the end of the OR schedule, and surgical staff must comply with individual decontamination procedure to prevent staff-to-patient transmission of microorganisms.^[42]

5. PREVENTION AFTER SURGERY

a) Temperature

The patient's temperature should be monitored and recorded at the time of admission to the recovery room and after that every 15 min and if it is recorded below 36°C, the patient should be actively warmed using forced air device.

b) Glucose level

There are no RCTs that have evaluated intensive glucose control to lower the preoperative average glucose (hemoglobinA1c) vs usual care before surgery, but postoperative hyperglycemia was associated with an increased risk of surgical site infections in patients with and without diabetes. As a result, strategies to prevent hyperglycemia to prevent surgical site infection are recommended in all major guidelines. SSI rates for general surgery patients increase dramatically to 10% at postoperative serum glucose levels of 140 mg/dL and higher. In colorectal surgery patients, 140 mg/dL was also the cutoff value for increased SSI rates.

c) Dressing of the surgical wound

An aseptic non-touch technique for changing or removing surgical wound dressings should be incorporated. Sterile saline is used for wound dressing up to 48 hours after surgery and patient should be advised that they may shower safely 28 hours after surgery. The surgical wounds that are healing by primary intention, the topical anti-microbial agents should not be used and the surgical wounds that are healing by secondary intention, an appropriate interactive dressing (Eusol and gauze, or moist cotton or mercuric antiseptic solution) should be used. The Eusol and gauze, or dextranomer or enzymatic treatments for debridement in the management of SSI. The patient should receive an antibiotic coverage as per the causative organism and results of microbiological tests and local resistance patterns when SSI is suspected by presence of cellulitis, either by a new infection or an infection caused by treatment failure.^[31]

Incisional negative pressure wound therapy can reduce the risk of SSI by promoting reducing fluid accumulation in the wounds, thereby accelerating primary wound healing.^[9]

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